

Metric Spaces | (Distance)

Def | (X, d) is a metric space if X is a set ($X \neq \emptyset$)

$$d: X \times X \rightarrow [0, \infty) \text{ s.t.}$$

$$1) \forall x, y \in X \quad d(x, y) = 0 \text{ iff } x = y$$

$$2) \forall x, y \in X \quad d(x, y) = d(y, x) \quad (\text{Symmetric})$$

$$3) \forall x, y, z \in X \quad d(x, z) \leq d(x, y) + d(y, z) \quad (\text{Triangle Inequality})$$

Difference w/ norm space

1) Norm spaces are vector spaces

2) $\|\cdot\|$ is from $X \rightarrow [0, \infty)$ s.t.,

$$\|a\vec{x}\| = |a| \|\vec{x}\|$$

$$3) \|\vec{x} + \vec{y}\| \leq \|\vec{x}\| + \|\vec{y}\|$$

If $(X, \|\cdot\|)$ is a norm space, then (X, d) is a metric space
with $d(x, y) = \|\vec{x} - \vec{y}\|$
 $\|\vec{x} - \vec{z}\| \leq \|\vec{x} - \vec{y}\| + \|\vec{y} - \vec{z}\|$

Examples:

Ex 1: $X = \mathbb{R}^d, d(x, y) = |x - y|$

$$|x - y| = \sqrt{\sum (x_i - y_i)^2}$$

Can verify triangle inequality with Cauchy-Schwarz

Ex 2: $X = \mathbb{R}^d$, $p \in (0, \infty]$

Define $\|x\|_p = \left(\sum |x_i|_p \right)^{1/p}$ if $p < \infty$

$$\|x\|_p = \max_{1 \leq i \leq d} |x_i|$$

Proof $(\mathbb{R}^d, \|\cdot\|_p)$ is a normed vector space iff $p \geq 1$

i.e. the fn $d(x, y) = \|x - y\|_p$ defines a metric on $\mathbb{R}^d$ iff $p \geq 1$

Sketch of a proof for $p \in [1, \infty)$:

NTS $\|x + y\|_p \leq \|x\|_p + \|y\|_p$

① Basic Calculus: $a, b > 0 \Rightarrow ab \leq \frac{a^p}{p} + \frac{b^q}{q}$ where $\frac{1}{p} + \frac{1}{q} = 1$
(Young's Inequality)

② Hölder's Inequality: $x, y \in \mathbb{R}$, $p \in [1, \infty]$

$$|x \cdot y| \leq |x|_p |y|_q \quad \frac{1}{p} + \frac{1}{q} = 1$$

③ $\forall x, y, z \in \mathbb{R}^d \quad |(x+y) \cdot z| \leq |x \cdot z| + |y \cdot z|$

$$\begin{aligned} &\stackrel{\text{Hölder}}{\leq} |x|_p |z|_q + |y|_p |z|_q \\ &= (|x|_p + |y|_p) |z|_q \end{aligned}$$

Choose z cleverly

Make $|(x+y) \cdot z| = \|x+y\|_p$

and make $|z|_q = 1$

Continuous Functions

Possible definitions:

$$\forall a \in \mathbb{R} \quad \lim_{x \rightarrow a} f(x) = f(a)$$

Def $f: \mathbb{R} \rightarrow \mathbb{R}$, $a \in \mathbb{R}$, $l \in \mathbb{R}$ we say

$$\lim_{x \rightarrow a} f(x) = l \quad \forall \varepsilon > 0 \exists \delta > 0 \text{ s.t. } \forall x \\ |x - a| < \delta \Rightarrow |f(x) - l| < \varepsilon$$

You can replace absolute value with any metric distance

Def (X, d_x) (Y, d_y)

$$f: X \rightarrow Y \quad a \in X, l \in Y$$

We say $\lim_{x \rightarrow a} f(x) = l$ if $\forall \varepsilon > 0 \exists \delta > 0$ s.t. $\forall x \in X$

$$0 < d_x(x, a) < \delta \Rightarrow d_y(f(x), l) < \varepsilon$$

Notation: $\mathcal{B}(x, r) = \{y \in Y, d(x, y) < r\}$ (Ball center x radius r)

$$\mathcal{B}^*(x, r) = \{y \in Y, 0 < d(x, y) < r\} = \mathcal{B}(x, r) - \{x\}$$

(Punctured Ball)

$$\lim_{x \rightarrow a} f(x) = l \Rightarrow \forall \varepsilon > 0 \exists \delta > 0 \text{ s.t. } x \in \mathcal{B}^*(a, \delta) \Rightarrow f(x) \in \mathcal{B}(l, \varepsilon)$$

Remark Sp. $X = \mathbb{Z}$ $f: X \rightarrow Y$
Euclidean distance

$$\forall a \in \mathbb{Z} \quad \forall l \in Y \quad \lim_{x \rightarrow a} f(x) = l \quad \text{Pf: } \forall \varepsilon > 0, \delta = \frac{1}{2}$$

Vacuously true

Prop $X = \mathbb{R}^d$ (Euclidean metric)

$f: X \rightarrow Y$ If $\lim_{x \rightarrow a} f(x) = l \wedge \lim_{x \rightarrow a} f(x) = m \Rightarrow l = m$

Pf

Let $\epsilon > 0$ be arbitrary

Know $\exists \delta_1$ s.t. $x \in \mathcal{B}^*(a, \delta_1) \Rightarrow f(x) \in \mathcal{B}(l, \epsilon)$
 $\exists \delta_2$ s.t. $x \in \mathcal{B}^*(a, \delta_2) \Rightarrow f(x) \in \mathcal{B}(m, \epsilon)$

Choose $\delta = \min \{\delta_1, \delta_2\}$ and $x \in \mathcal{B}^*(a, \delta) \neq \emptyset$

Then $d(f(x), l) < \epsilon \wedge d(f(x), m) < \epsilon$

$\Rightarrow d(l, m) \leq d(f(x), l) + d(f(x), m) < 2\epsilon$

So $l = m$

We need this non-emptiness property

Def

$a \in X$ is an isolated point if $\exists \delta > 0$ s.t. $\mathcal{B}^*(a, \delta) = \emptyset$

Prop If $a \in X$ is not an isolated point,

then if $\lim_{x \rightarrow a} f(x)$ exists it is unique,

$f: X \rightarrow Y$ (X, Y metric)

Proof: Say $Y = \mathbb{R}$ (Euclidean Metric)

Say $f, g: X \rightarrow \mathbb{R}$ $\lim_{x \rightarrow a} f(x) = l$ $\lim_{x \rightarrow a} g(x) = m$

Want to conclude: $\lim_{x \rightarrow a} f(x)g(x) = l+m$

PEI let $\epsilon > 0$ be arbitrary

Choose some δ s.t. $x \in \mathcal{B}^*(a, \delta) \Rightarrow f(x)g(x) \in \mathcal{B}(l+m, \epsilon)$

$$\Leftrightarrow |f(x)g(x) - (l+m)| < \epsilon$$

So $\exists \delta_1 > 0$ s.t. $x \in \mathcal{B}^*(a, \delta_1) \Rightarrow |f(x) - l| < \frac{\epsilon}{2}$

Also, $\exists \delta_2 > 0$ s.t. $x \in \mathcal{B}^*(a, \delta_2) \Rightarrow |g(x) - m| < \frac{\epsilon}{2}$

So, let $\delta = \min(\delta_1, \delta_2)$

$x \in \mathcal{B}^*(a, \delta) \Rightarrow x \in \mathcal{B}^*(a, \delta_1) \wedge x \in \mathcal{B}^*(a, \delta_2) \Rightarrow$

$$|f(x)g(x) - l + g(x) - m| \leq |f(x) - l| + |g(x) - m| < \epsilon$$

Wrong proof of $\lim_{x \rightarrow a} f(x)g(x) = lm$

let $\epsilon > 0$ $d(f(x)g(x), lm) < \epsilon \Leftrightarrow |f(x)g(x) - lm| < \epsilon$

Can make $|f(x) - l|$ arbitrarily small

and $|g(x) - m|$ arbitrarily small

$$|f(x)g(x) - lm| = |f(x)g(x) - lg(x) + lg(x) - lm| \leq |g(x)| |f(x) - l| + |l| |g(x) - m|$$

$$\exists \delta_1 > 0 \text{ s.t. } |g(x) - m| < \frac{\epsilon}{2(l+1)}$$

$$\exists \delta_2 > 0 \text{ s.t. } |f(x) - l| < \frac{\epsilon}{2|g(x)|+1} \leftarrow \text{Wrong}$$

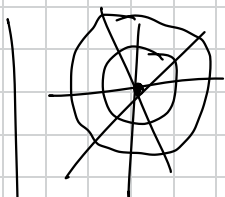
Fix: Get an upper bound for $|g(x)|$

Choose $\epsilon = 1$. $\exists \delta_2 > 0$ s.t. $x \in \mathcal{B}^*(a, \delta_2) \Rightarrow |g(x) - m| < 1$

$$|g(x)| < |m| + 1$$

Choose δ_3 s.t. $x \in \mathcal{B}^*(a, \delta_3) \Rightarrow |f(x) - l| < \frac{\epsilon}{2(|m|+1)}$

$$|g(x)| |f(x) - l| < \epsilon$$



Reduce to just lines

Sp. $f: X \rightarrow \mathbb{R}^d$ $a \in X$

$$\lim_{x \rightarrow a} f(x) = l \iff \forall i \in \{1, \dots, d\} \quad \lim_{x \rightarrow a} f(x)_i = l_i$$

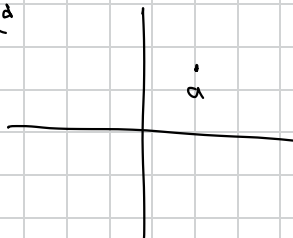
Pf $\Leftarrow$

Find δ_i s.t. $x \in \mathcal{B}^*(x, \delta_i) \Rightarrow |f(x)_i - l_i| < \frac{\epsilon}{\sqrt{d}}$

Let $\delta = \min \delta_i$ So $|f(x)_i - l_i| < \frac{\epsilon}{\sqrt{d}} \quad \forall i$

$$\text{So } |f(x) - l| < \epsilon$$

Sp: $f: \mathbb{R}^d \rightarrow Y$ $a \in \mathbb{R}^d$



If $\lim_{x \rightarrow a} f(x) = l$ then $\forall v \in \mathbb{R}^d \neq 0$

$$\lim_{t \rightarrow 0} f(a+tv) = l$$

(1D limit)

Pf (Scratch) Want $d(f(a+tv), l) < \epsilon$ when $0 < |t| < \delta$

Know $\exists \delta$ s.t. $0 < |x-a| < \delta \Rightarrow d(f(x), l) < \epsilon$

Choose δ to where $|a+tv-a| < \delta$,
 $|t||v| < \delta$,

Let $\epsilon > 0$

$\exists \delta_1 > 0$ s.t. $0 < |x-a| < \delta_1 \Rightarrow d(f(x), l) < \epsilon$

Choose $\delta = \frac{\delta_1}{|v|}$ Sps. $0 < |t| < \delta$

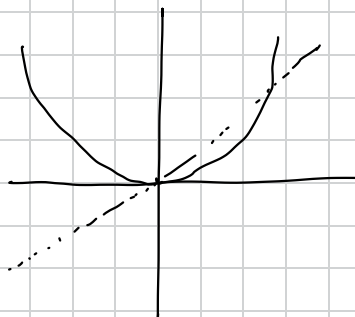
$$\Rightarrow 0 < |a+tv-a| = |t||v| < \delta|v| = \delta_1$$

$$\Rightarrow d(f(a+tv), l) < \epsilon$$

Converse is false

$$f: \mathbb{R}^2 \rightarrow \mathbb{R} \quad \lim_{t \rightarrow 0} f(a+tv) = 0 \quad \forall v \in \mathbb{R}^2, v \neq 0$$

But: $\lim_{x \rightarrow a} f(x) \neq 0$



1 on parabola
0 everywhere else

Let $f(x) = \begin{cases} 1 & \text{if } x_2 = x_1^2 \\ 0 & \text{if } x_2 \neq x_1^2 \end{cases}$ Claim:

$$\textcircled{1} \lim_{t \rightarrow 0} f(0+tv) = 0 \quad \forall v \in \mathbb{R}^2, v \neq 0$$

$$\textcircled{2} \lim_{x \rightarrow 0} f(x) = \text{DNE}$$

Limits of ∞

$$x \in \mathbb{R} \quad f: \mathbb{R} \rightarrow Y$$
$$\lim_{x \rightarrow \infty} f(x) = l \quad \text{if } \forall \varepsilon > 0, \exists R \in \mathbb{R} \text{ s.t. } x > R \Rightarrow d(f(x), l) < \varepsilon$$

$[-\infty, \infty] = \mathbb{Z}$ Find a metric

Continuity

We say f is continuous at a if

$$\lim_{x \rightarrow a} f(x) = f(a)$$

We say $f: X \rightarrow Y$ is continuous if $\forall a \in X$,
 f is continuous at a

f cts at $a \iff \forall \varepsilon > 0, \exists \delta$ s.t. $x \in \mathcal{B}(a, \delta) \Rightarrow d(f(x), f(a)) < \varepsilon$

↖ No puncture

If $f, g: X \rightarrow \mathbb{R}^d$ are cts, then $f+g$ is cts.

Pf

$$\lim_{x \rightarrow a} f(x) + g(x) = \lim_{x \rightarrow a} f(x) + \lim_{x \rightarrow a} g(x) = f(a) + g(a)$$

Thm 1 $X \xrightarrow{g} Y \xrightarrow{f} Z$

Consider $f(g(x)): X \rightarrow Z$

If g is cts at a , f is cts at $g(a)$

then $f \circ g$ is cts at a

Pf Pick $\epsilon > 0$ WTS: $d(x, a) < \delta \Rightarrow d(f \circ g(x), f \circ g(a)) < \epsilon$

f cts. $\Rightarrow \exists \delta_1 > 0$ s.t. $d(y, g(a)) < \delta_1 \Rightarrow d(f(y), f(g(a))) < \epsilon$

where g cts. $\Rightarrow \exists \delta_2 > 0$ s.t. $d(x, a) < \delta_2 \Rightarrow d(g(x), g(a)) < \delta_1$

$\therefore d(x, a) < \delta_2 \Rightarrow d(f(g(x)), f(g(a))) < \epsilon$

Prop 1 $T: \mathbb{R}^m \rightarrow \mathbb{R}^n$ linear

$\Rightarrow T$ is continuous

Pf Sketch | Pick $\epsilon > 0$ $a \in \mathbb{R}^m$

Want $|x - a| < \delta \Rightarrow |Tx - Ta| < \epsilon$
 $\Leftrightarrow |T(x - a)| < \epsilon$

$|Tv| \leq \|T\| |v|$

$\uparrow$
operator norm

Claim: $\exists c \in \mathbb{R}$ s.t. $\forall v \in \mathbb{R}^m$

$|Tv| \leq c |v|$

Choose c to be (max abs value of the entries in T) $\cdot n$

Given the claim

Pick $\epsilon > 0$ Choose $\delta = \frac{\epsilon}{c+1}$

Know that f, g lts $\Rightarrow f+g$ lts

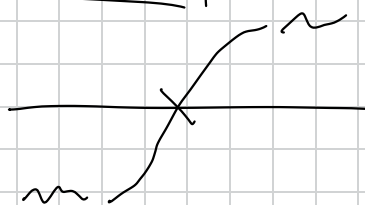
Alt proof:

f, g is lts

Compos $\left\{ \begin{array}{l} \textcircled{1} x \rightarrow (f(x), g(x)) \text{ is lts} \\ \textcircled{2} (y, z) \rightarrow y+z \text{ is lts} \\ \quad \text{(linear)} \end{array} \right.$

$x \rightarrow f(x) + g(x)$ is lts

Real Numbers



IVT

Binary search for a root
Take limit

Def The set of real numbers is a complete ordered field

Def A field is a set F along w/ 2 binary operations $+$ and $\cdot$ satisfying the "usual" axioms

Def An ordered field is a field F along with an order " $<$ " s.t.

① $\forall x, y \in F$ either $x < y$, $y < x$, or $x = y$

② $x < y$ and $0 < c \Rightarrow cx < cy$

③ $x < y$ and $c \in F \Rightarrow x + c < y + c$

$a < b \wedge b < c \Rightarrow a < c$

We define $x > y$ if $y < x$, and $\leq \geq$ similarly

Ordered field examples

$\mathbb{Q}$ are an ordered field

$F = \{a + b\sqrt{2} \mid a, b \in \mathbb{Q}\}$ is an ordered field

$\mathbb{C}$ is a field but not ordered

Def | F ordered field

$A \subseteq F$ we say $u \in F$ is an upper bound of A if

$$\forall a \in A, a \leq u$$

Def | We say β is the supremum of A if

1. β is an upper bound of A

2. u upper bound $\Rightarrow \beta \leq u$

Def | We say α is the infimum of A if

1. α is a lower bound of A

2. l lower bound $\Rightarrow \alpha \geq l$

We say a field is order complete if every non-empty subset bounded above has a supremum.

E.g.: $\mathbb{Q} \rightarrow$ order complete? No!

$$A = \{q \in \mathbb{Q} \mid q^2 < 2\} \quad \text{Expect } \sup(A) = \sqrt{2} \notin \mathbb{Q}$$

$\sup(A)$ is unique

$$\beta_1, \beta_2 \text{ are two suprema} \quad \beta_1 \leq \beta_2, \beta_2 \leq \beta_1 \Rightarrow \beta_1 = \beta_2$$

Thm | There exists a complete ordered field, which is unique up to isomorphism

Def $\mathbb{R}$ is a complete ordered field

Prop | Natural numbers is not bounded above.

PH AFSOC α is an U.B of $\mathbb{N}$

OC: $\exists \beta \in \mathbb{R}$ s.t. $\beta = \sup(\mathbb{N})$

Consider $\beta - 1$, this is not an upper bound. of $\mathbb{N}$

$$\exists n \in \mathbb{N} \rightarrow \beta - 1 < n \Rightarrow \beta < n + 1 \in \mathbb{N}$$

~~X~~

Prop |

1. Any real lies between two consecutive integers
2. Any real number has a decimal expansion
3. Between any 2 reals there is a rational and an irrational

Sequences

Def | A function $\alpha: \mathbb{N} \rightarrow Y$ is called a sequence

same metric space

Notation: $(a_0, a_1, \dots) \leftarrow$ sequence

Also $(a_n)_{n \in \mathbb{N}}$ notation for a seq or even just (a_n)

Def

We say $a_n \rightarrow l$
("converges to l ")

$$\lim_{n \rightarrow \infty} a_n = l \iff \forall \varepsilon > 0, \exists N \in \mathbb{N} \text{ s.t. } \forall n > N \quad d(a_n, l) < \varepsilon$$

E.g. The seq $\frac{1}{n} \rightarrow 0$ as $n \rightarrow \infty$

Pf | Let $\varepsilon > 0$ be const. Choose $N \in \mathbb{N}$ s.t. $N > \frac{1}{\varepsilon}$ (Archimedean Principle)

$$n > N \Rightarrow n > \frac{1}{\varepsilon} \Rightarrow \frac{1}{n} < \varepsilon \Rightarrow d(a_n, 0) < \varepsilon$$

Limit Theorems for functions (e.g. $\lim_{x \rightarrow a} f(x) + g(x) = \lim_{x \rightarrow a} f(x) + \lim_{x \rightarrow a} g(x)$) are also true for sequences.

Thm | $f: X \rightarrow Y$ $a \in X$

f is cts. at $a \iff$ for every sequence (a_n) in X ,
whenever $a_n \xrightarrow{n \rightarrow \infty} a$ we have $f(a_n) \xrightarrow{n \rightarrow \infty} f(a)$

Pf | $\Leftarrow \forall$ seq (a_n) s.t. $a_n \xrightarrow{n \rightarrow \infty} a$ we have $f(a_n) \xrightarrow{n \rightarrow \infty} f(a)$

$$\text{NTS: } \lim_{x \rightarrow a} f(x) = f(a)$$

$$\text{AFSOC } \lim_{x \rightarrow a} f(x) \neq f(a)$$

$$\Rightarrow \exists \varepsilon > 0, \forall \delta > 0, \exists x \in X \text{ s.t. } d(x, a) < \delta \text{ \& } d(f(x), f(a)) \geq \varepsilon$$

$$\forall n \in \mathbb{N}, \text{ choose } \delta = \frac{1}{n} \Rightarrow \exists x_n \in X \text{ s.t. } d(x_n, a) < \frac{1}{n} \text{ \& } d(f(x_n), f(a)) \geq \varepsilon$$

Note: ① $x_n \xrightarrow{n \rightarrow \infty} a$ ($\because d(x_n, a) < \frac{1}{n}$)

② $f(x_n) \not\xrightarrow{n \rightarrow \infty} f(a)$ ($\because$ for the ε above, $d(f(x_n), f(a)) \geq \varepsilon \forall n \in \mathbb{N}$)
✗

Def | $a: \mathbb{N} \rightarrow \mathbb{R}$ is called increasing if $\forall n \in \mathbb{N}, a_{n+1} \geq a_n$

strictly increasing if $a_{n+1} > a_n$

Thm | (a_n) is increasing and bounded then (a_n) is convergent

bounded (a_n) if $\{a_n | n \in \mathbb{N}\}$ is bounded

PF | Let $l = \sup \{a_n | n \in \mathbb{N}\}$
Claim: $(a_n) \xrightarrow{n \rightarrow \infty} l$

Let $\varepsilon > 0$ l supremum $\Rightarrow l - \varepsilon$ is not an upper bound
 $\Rightarrow \exists N$ s.t. $a_n > l - \varepsilon$

$$\forall n \geq N \quad \underset{\substack{\text{increasing} \\ \uparrow \\ (l \text{ supremum})}}{l - \varepsilon < a_n \leq l} \Rightarrow d(a_n, l) < \varepsilon$$

Int Value Thm

$f: [a, b] \rightarrow \mathbb{R}$ cts $f(a) < 0, f(b) > 0 \Rightarrow \exists x \in (a, b)$ s.t. $f(x) = 0$



$$\begin{aligned} (a_n) &\rightarrow r & (b_n) &\rightarrow r \\ a_n &\leq r & b_n &\geq r \end{aligned}$$

lim sup & lim inf

$$a: \mathbb{N} \rightarrow \mathbb{R}$$

$$\alpha_k = \inf \{a_n \mid n \geq k\}$$

$$\beta_k = \sup \{a_n \mid n \geq k\}$$

By convention all these $\pm \infty$ if not correspondingly bounded

There are monotonic sequences $\alpha_{k+1} \geq \alpha_k, \beta_{k+1} \leq \beta_k$

Suppose the sequence (a_n) is bounded

$$(a_n) \text{ bounded} \Rightarrow \alpha_k \leq \beta_1 < \infty \text{ as } (a_n) \text{ is bounded}$$

$$\Rightarrow \beta_k \geq \alpha_k$$

Both (α_k) and (β_k)
are convergent sequences

Def

$$\liminf_{n \rightarrow \infty} a_n = \lim_{k \rightarrow \infty} \alpha_k = \sup \{\alpha_k : k \in \mathbb{N}\}$$

$$\limsup_{n \rightarrow \infty} a_n = \lim_{k \rightarrow \infty} \beta_k = \inf \{\beta_k : k \in \mathbb{N}\}$$

$$\liminf_{n \rightarrow \infty} a_n \leq \lim_{n \rightarrow \infty} a_n \leq \limsup_{n \rightarrow \infty} a_n$$

Thm

$a: \mathbb{N} \rightarrow \mathbb{R}$ is a bounded sequence

$$\textcircled{1} \liminf_{n \rightarrow \infty} a_n \leq \limsup_{n \rightarrow \infty} a_n$$

$$\textcircled{2} \liminf_{n \rightarrow \infty} a_n = \limsup_{n \rightarrow \infty} a_n \iff (a_n) \text{ is convergent}$$

$$\text{and } \liminf = \lim = \limsup$$

Pf $\Rightarrow$ Suppose $\liminf_{n \rightarrow \infty} a_n = \limsup_{n \rightarrow \infty} a_n$

$$\text{NTS: } \liminf_{n \rightarrow \infty} a_n = \limsup_{n \rightarrow \infty} a_n = \dots$$

$$\text{Let } \alpha_k = \inf \{a_n : n \geq k\} \quad l = \sup \{\alpha_k\} = \inf \{\beta_k\}$$

$$\beta_k = \sup \{a_n : n \geq k\}$$

$$\text{Let } \varepsilon > 0$$

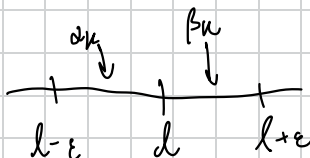
$$l - \varepsilon \text{ is not an u.b. of } \{\alpha_k\}$$

So there exists some $k_1 \in \mathbb{N}$ s.t.

$$\alpha_{k_1} > l - \varepsilon$$

Similarly, $\exists k_2 \in \mathbb{N}$ s.t. $\beta_{k_2} < l + \varepsilon$

$$\text{Choose } N = \max \{k_1, k_2\} \quad \forall n \geq N, l - \varepsilon < \alpha_{k_1} \leq a_n \leq \beta_{k_2} < l + \varepsilon$$



Def | Y some metric space

$$\alpha: \mathbb{N} \rightarrow Y$$

We say the sequence (α_n) is Cauchy if $\forall \varepsilon > 0$

$$\exists N \in \mathbb{N} \text{ s.t. } m, n \geq N \implies d(\alpha_m, \alpha_n) < \varepsilon$$

Cauchy Sequences need not be convergent ($\mathbb{Q}$)

Convergent $\rightarrow$ Cauchy

Prop | Any Convergent sequence is Cauchy

$$\forall \varepsilon > 0 \quad \exists N \in \mathbb{N} \text{ s.t. } n \geq N \implies d(\alpha_n, l) < \frac{\varepsilon}{2}$$

Now if $n, m \geq N$ then

$$d(\alpha_n, \alpha_m) \leq d(\alpha_n, l) + d(l, \alpha_m) < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon$$

Prop | Cauchy $\not\Rightarrow$ Convergent

Thm | $X = \mathbb{R}^d$, Cauchy $\Rightarrow$ Convergent

PF) ① Case I $d = 1$

$$\text{Will show } \overline{\lim}_{n \rightarrow \infty} \alpha_n = \lim_{n \rightarrow \infty} \alpha_n$$

$$\text{Choose } \varepsilon = 1 \quad \exists N \in \mathbb{N} \text{ s.t. } m, n \geq N \implies |\alpha_m - \alpha_n| \leq 1 \\ \implies |\alpha_m| \leq |\alpha_n| + 1$$

$$\text{Let } M = \max \{|\alpha_1|, \dots, |\alpha_N|, |\alpha_N| + 1\} \quad \forall n \in \mathbb{N} \quad |\alpha_n| \leq M$$

Thus, bounded

② Pick any $\varepsilon > 0$

$$\alpha_k = \inf \{\alpha_n : n \geq k\}$$

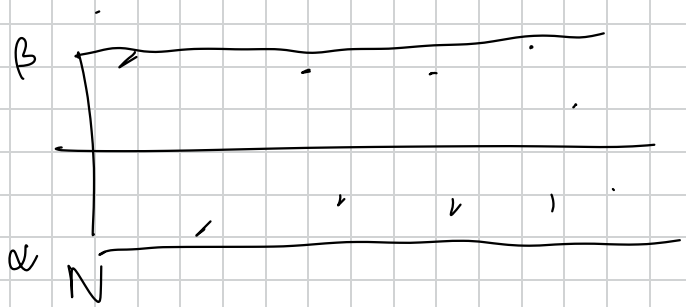
$$\beta_k = \sup \{\alpha_n : n \geq k\}$$

$$\exists N \text{ s.t. } m, n \geq N \implies |\alpha_m - \alpha_n| < \varepsilon$$

$$\implies \beta_N - \alpha_N \leq \varepsilon \quad (\text{Needs checking})$$

$$\Rightarrow 0 \leq \overline{\lim} a_n - \underline{\lim} a_n \leq \varepsilon$$

$$\overline{\lim} a_n = \underline{\lim} a_n$$



Recall (a_n) is a seq. in $\mathbb{R}^d$

Def

$$s_n = \sum_{k=1}^n a_k \quad (\text{partial sum})$$

Def We say $\sum_{n=1}^{\infty} a_n$ converges if $\lim_{n \rightarrow \infty} s_n$ exists

$$\sum_{n=1}^{\infty} a_n = \lim_{n \rightarrow \infty} s_n$$

Def The series $\sum_{n=1}^{\infty} a_n$ converges absolutely if the series $\sum_{n=1}^{\infty} |a_n|$ converges

Note: this $\sum_{n=1}^{\infty} |a_n|$ is increasing, so converges iff it is bounded

Convention We say $\sum_{n=1}^{\infty} |a_n| < \infty$

Thm | Absolute Convergence $\Rightarrow$ Convergence of a series
(But not conversely)

PF | Let $s_n = \sum_{i=1}^n |a_i|$
Let $t_n = \sum_{i=1}^n a_i$

WTS: (t_n) is convergent

Show that it is Cauchy
 $\forall \epsilon > 0$, will show $|t_n - t_m| < \epsilon \quad \forall n, m$

Note: (s_n) cgt $\Rightarrow (s_n)$ Cauchy $\Rightarrow \exists N \in \mathbb{N}$ s.t. $\forall n, m \geq N$
 $|s_n - s_m| < \epsilon$

WLOG: $m \leq n$
 $|s_n - s_m| = \sum_{i=m+1}^n |a_i|$
 $|t_n - t_m| = \left| \sum_{i=m+1}^n a_i \right| \leq \sum_{i=m+1}^n |a_i| < \epsilon$

$\Rightarrow (t_n)$ Cauchy $\Rightarrow (t_n)$ cgt

Thm | Comparison Test

Say $(a_n), (b_n)$ two sequences with $|a_n| \leq b_n$

If $\sum_{n=1}^{\infty} b_n$ is cgt, then $\sum_{n=1}^{\infty} a_n$ is absolutely cgt

$\sum_{n=1}^{\infty} |a_n|$ is bounded by $\sum_{n=1}^{\infty} b_n$

Thm 1

$\sigma: \mathbb{N} \rightarrow \mathbb{N}$ a bijection

Suppose $\sum a_n$ is convergent

Q: Is $\sum a_{\sigma(n)} = \sum a_n$?

If the series $\sum_{n=1}^{\infty} |a_n| < \infty$ then $\sum_{n=1}^{\infty} a_{\sigma(n)}$ is cgt

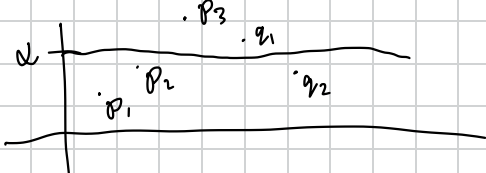
$$\& \sum_{n=1}^{\infty} a_{\sigma(n)} = \sum_{n=1}^{\infty} a_n$$

Thm 1 Say $\sum_{n=1}^{\infty} a_n$ is conditionally cgt (i.e. $\sum a_n$ is cgt but $\sum |a_n|$ is not)

Then, $\forall \alpha \in \mathbb{R}^d \exists \sigma \in \mathbb{N} \rightarrow \mathbb{N}$ s.t. $\sum a_{\sigma(n)} = \alpha$

Pf Idea | $p_n = \max(a_n, 0)$ $q_n = -\min(a_n, 0)$

① $\sum p_n, \sum q_n$ are NOT cgt (bc a_n is not absolutely cgt)

②  Keep on taking until crossing

a_n must converge to 0

$$\sum_{i=1}^n a_i \rightarrow \sigma \quad \sum_{i=1}^{n+1} a_i \rightarrow \sigma \quad a_{n+1} = \sum_{i=1}^{n+1} a_i - \sum_{i=1}^n a_i = 0$$

Theorem 18

$$P(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_0$$
$$= a_n x^n \left(1 + \frac{a_{n-1}}{a_n} x^{-1} + \dots + \frac{a_0}{a_n} x^{-n} \right)$$

We know $a_n \neq 0$, n is odd WLOG let $a_n > 0$ (proof is mirrored if the other case).

$$\lim_{x \rightarrow \infty} P(x) = \infty \quad \lim_{x \rightarrow -\infty} P(x) = -\infty$$

Thus, $\exists a$ s.t. $P(a) > 0$ and $\exists b$ s.t. $P(b) < 0$

So by IVT, $\exists x \in [a, b]$ s.t. $P(x) = 0$

Subsequences | $a: \mathbb{N} \rightarrow X$ a seq

(a_m) a seq in X

$(n_k)_{k \in \mathbb{N}}$ a strictly inc seq in $\mathbb{N}$

We say $(a_{n_k})_{k \in \mathbb{N}}$ is a subsequence of $(a_n)_{n \in \mathbb{N}}$

$(a_m) \xrightarrow{m \rightarrow \infty} l$ $(a_{n_k}) \rightarrow$ any subsequence

Then $(a_{n_k}) \xrightarrow{k \rightarrow \infty} l$

PF | Let $\epsilon > 0$ be arbitrary

$$(a_n) \xrightarrow{n \rightarrow \infty} l \Rightarrow \exists N \text{ s.t. } n \geq N \Rightarrow d(a_n, l) < \epsilon$$

Choose $K = N$

Assume $k \geq K$ Since (n_k) is strictly increasing
 $n_k \geq k \geq K = N$

$$d(a_{n_k}, l) < \epsilon$$

Eg $a_m = (-1)^m$ Does not converge

$$(a_{2m+1}) \xrightarrow{m \rightarrow \infty} -1 \quad (a_{2m}) \xrightarrow{m \rightarrow \infty} 1$$

So a_m does not converge

Does any seq. have a cgt. subseq.

NO: $a_n = n$

Thm (Bolzano - Weierstrass)

Any bounded seq. in $\mathbb{R}^d$ has a cgt. subsequence

Def A set $K \subseteq X$ is called (sequentially) compact if any seq. in K has a cgt. subseq. in K

i.e. If (a_m) is a seq. & $a_m \in K \forall m$
Then $\exists$ a subseq. (a_{n_k}) s.t. $\lim_{k \rightarrow \infty} a_{n_k}$ exists & $\in K$

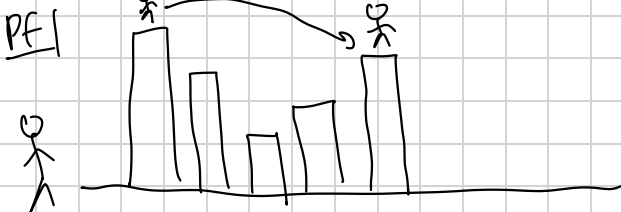
Is $\emptyset$ seq. cpt? Yes (vacuously)

Is $\mathbb{R}$ seq. cpt? No ($a_n = n$)

If $K \subseteq X$ is finite? Yes (one term repeats infinitely)

Lemma (Rising Sun)

If (a_n) is a seq. in $\mathbb{R}$, then (a_n) has a monotone subsequence



Choose $n_1 = \min\{n \mid a_n \geq a_k \forall k \geq n\}$ (if possible)

$n_2 = \min\{n > n_1 \mid a_n \geq a_k \forall k \geq n\}$ (if possible)

$n_{k+1} = \min\{n > n_k \mid a_n \geq a_k \forall k \geq n\}$ (if possible)

If $\forall k \in \mathbb{N}$ the choice of n_k exists

then (a_{n_k}) is a monotone seq. subsequence

If $\exists k \in \mathbb{N}$ s.t. the choice of n_k DNE

Then let $m_1 = n_{k-1} + 1 \exists m_2 > m_1$ s.t. $a_{m_2} > a_{m_1}$

$\exists m_3 > m_2$ s.t. $a_{m_3} > a_{m_2}$
 $\vdots$

Bolzano-Weierstrass

$K \subseteq \mathbb{R}^d$ is sequentially compact iff K is both closed and bounded

Remark: Say $K \subseteq X$ is seq. compact $\Rightarrow K$ closed and bounded

(Any metric space)

Prop 1 K seq comp. $\Rightarrow K$ closed & K bounded

$B \subseteq X$ is bounded if $\exists x_0 \in X \wedge r < \infty$ st. $B \subseteq B(x_0, r)$

Pf idea | If not closed, construct a sequence that converges to a point outside
If not bounded, construct a sequence that blows up to ∞

$K \subseteq \mathbb{R}^d$ is closed and bounded $\Rightarrow K$ is seq compact

Pf | Case I: $d=1$

Assume (a_n) is a seq in K
 $\exists$ monotone subseq. (a_{n_k})

Thus monotone + bounded $\Rightarrow$ convergent

Since K is closed, any limit in K is in K
 $\lim_{k \rightarrow \infty} a_{n_k} \in K$

Case II: $d \geq 2$

Look at each coordinate

(a_n) is a seq in K let $a_n^i = i^{\text{th}}$ coordinate of a_n

Note: (a_n^1) is a bounded sequence in $\mathbb{R}$

Thus, $\exists$ a g.t. subseq. $(a_{n_k}^1)$

Look at $(a_{n_k}^2)$ is a bounded sequence in $\mathbb{R}$

Thus $\exists$ a g.t. subseq. $(a_{n_{k_l}}^2)$

We continue this process until we get our desired result

Thm | Say $f: X \rightarrow Y$ is cts

Say $K \subseteq X$ is seq. compact

Then $f(K) \subseteq Y$ is seq. compact

Pf |

Say (y_n) is a seq. in $f(K)$

Take the pre-images (x_n) s.t. $f(x_n) = y_n$

Take the subsequence of (x_{n_k}) s.t. $\lim_{k \rightarrow \infty} x_{n_k}$ exists & $\in K$

Say $(x_{n_k}) \rightarrow \alpha \in K$ f cts $\Rightarrow (f(x_{n_k})) \xrightarrow{k \rightarrow \infty} f(\alpha) \Rightarrow (y_{n_k}) \rightarrow f(\alpha) \in f(K)$

Corollary | (Extreme Value Theorem)

Say K cpt & $f: K \rightarrow \mathbb{R}$ is cts

then $\exists m, M \in \mathbb{R}$ s.t. $\forall x \in K, f(m) \leq f(x) \leq f(M)$

Corollary | $f: [a, b] \rightarrow \mathbb{R}$ (same conclusion),

Pf | f cts. K cpt. $\Rightarrow f(K)$ seq. compact $\Rightarrow$ bounded and closed

Take sup and inf of which are in $f(K)$ since it's closed

$\Rightarrow \exists m, n \in K$ s.t. $f(m) = \sup f(K), f(n) = \inf f(K)$

Derivatives | $f: \mathbb{R} \rightarrow \mathbb{R}$

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

We say f is differentiable if this limit exists

Thm | f diff at $a \Rightarrow f$ cts at a

$$\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} \left(\frac{f(x) - f(a)}{x - a} (x - a) + f(a) \right) = f'(a) \cdot 0 + f(a) = f(a)$$

Thm | Chain Rule

g is diff at a , f is diff at $g(a)$

Then $f \circ g$ is diff at a and $(f \circ g)'(a) = f'(g(a)) \cdot g'(a)$

Scratch | We need to check

$$\lim_{x \rightarrow a} \frac{(f \circ g)(x) - (f \circ g)(a)}{x - a} = \lim_{x \rightarrow a} \underbrace{\frac{(f \circ g)(x) - (f \circ g)(a)}{g(x) - g(a)}}_{x \rightarrow a, f'(g(a))} \cdot \underbrace{\frac{g(x) - g(a)}{x - a}}_{g'(a)}$$

If $g(x) = g(a)$ original limit is 0 and $g'(a)$ would also be 0

Clean Proof | Define

$$G(x) = \begin{cases} \frac{g(x) - g(a)}{x - a} & \text{if } x \neq a \\ g'(a) & \text{if } x = a \end{cases}$$

g diff at $a \rightarrow G$ cts at a

f diff at $g(a) \rightarrow F$ cts at a

Let
$$F(y) = \begin{cases} \frac{f(y) - f(g(a))}{y - g(a)} & y \neq g(a) \\ f'(g(a)) & y = g(a) \end{cases}$$

Note: $\frac{f(g(x)) - f(g(a))}{x - a} = F(g(x)) - G(x)$

Since this is cts, just use limit rules

$$\lim_{x \rightarrow a} \frac{f(g(x)) - f(g(a))}{x - a} = \lim_{x \rightarrow a} F(g(x)) \cdot \lim_{x \rightarrow a} G(x) \quad \text{Continuity of composites}$$

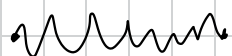
$$= F(g(a)) \cdot g'(a) = f'(g(a)) \cdot g'(a)$$

Thm (Lagrange Mean Value Theorem)

$f: [a, b] \rightarrow \mathbb{R}$ Assume f is cts. on $[a, b]$
 f is diff. on (a, b)

Then $\exists x \in (a, b)$ s.t. $f'(x) = \frac{f(b) - f(a)}{b - a}$

Roll's

$f(a) = f(b) = 0$  Produce one point where derivative is 0

Use EVT to find maxima

You can prove Lagrange MVT with the function being the distance from secant.

Thm (Cauchy Mean Value Theorem)

$f, g: [a, b] \rightarrow \mathbb{R}$ Assume f, g cts on $[a, b]$
 f, g diff on (a, b)

then $\exists x \in (a, b)$ s.t.

$$f'(x)[g(b) - g(a)] = g'(x)[f(b) - f(a)]$$

To get Lagrange let $g(x) = x$

Intuition: 2 people running a race. If one ran twice the distance of the other, there must have been a point in which they were running twice as fast.

Lemma If f is diff at a and obtains a local maxima at a then $f'(a) = 0$

$f: U \rightarrow \mathbb{R}$

Pf Say f attains a local maxima at a

$$\exists r > 0 \text{ s.t. } x \in \mathcal{B}(a, r), f(a) \geq f(x)$$

$$\forall \delta \in (0, r) \exists x_1, x_2 \in \mathcal{B}^*(a, \delta) \text{ s.t. } \frac{f(x_1) - f(a)}{x_1 - a} \geq 0$$

$$\text{and } \frac{f(x_2) - f(a)}{x_2 - a} \leq 0$$

Since local max, pick one to the left one to the right
(numerator is always negative),

$$\text{Thus, } \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} \text{ exists } \Rightarrow f'(a) = 0$$

Pf of Cauchy MVT

$$\text{Let } h(x) = (g(b) - g(a))(f(x) - f(a)) - (f(b) - f(a))(g(x) - g(a))$$

$$h(a) = 0$$

$$h(b) = 0$$

h is differentiable Find a point where $h' = 0$

(we can get this from EVT).

Case 1: Either of the min or max will be in (a, b)

Just take them and $h' = 0$

Case 2: Both extrema are not in (a, b)

That means 0 everywhere, grab any point

Prop 1 | Let $I \subseteq \mathbb{R}$ be an interval

$f: I \rightarrow \mathbb{R}$ be cts on I

$f: I \rightarrow \mathbb{R}$ be diff on $\overset{\circ}{I}$

If $f' > 0$ on $\overset{\circ}{I}$ then f is strictly increasing
(Converse not necessarily true i.e. x^3)

PF | f strictly increasing means whenever $x < y$, $f(x) < f(y)$

Choose $x < y \in I$ MVT $\Rightarrow \exists \xi \in (x, y)$ s.t. $f(y) - f(x) = f'(\xi)(y - x)$
 $\uparrow$ positive $\uparrow$ positive

So $f(y) - f(x) > 0$

Prop 1 (L'Hopital's Rule)

f, g : diff. functions, $a \in \mathbb{R}$ $g' \neq 0$ close to a

If $\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} g(x) = 0$ & $\lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$ exists

Then

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$$

PF (Scratch)

Recall Cauchy MVT: $\frac{f'(\xi)}{g'(\xi)} = \frac{f(b) - f(a)}{g(b) - g(a)}$

$$\frac{f(x)}{g(x)} = \frac{f(x) - f(a)}{g(x) - g(a)}$$

Redefine these functions so
 $f(a) = 0$ and $g(a) = 0$
(Doesn't matter cause we're not considering a)

PE1 Let $\varepsilon > 0$ Define $f(a) = g(a) = 0$ let $\lim_{x \rightarrow a} \frac{f'(x)}{g'(x)} = L$

Note: f, g are cts at a & diff. except possibly at a

Pick $\delta > 0$ st. $0 < |x - a| < \delta \rightarrow \left| \frac{f'(x)}{g'(x)} - L \right| < \varepsilon$

Pick any y s.t. $0 < |y - a| < \delta$

By Cauchy MVT,

$$\exists x \text{ between } a \text{ and } y \text{ st. } \frac{f(y)}{g(y)} = \frac{f(y) - f(a)}{g(y) - g(a)} = \frac{f'(x)}{g'(x)}$$

$$\text{So } 0 < |x - a| < \delta \Rightarrow \left| \frac{f'(x)}{g'(x)} - L \right| < \varepsilon$$

$$\text{So } \left| \frac{f(y)}{g(y)} - L \right| < \varepsilon \quad \text{So } \lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$$

$f \rightarrow$ diff. at a

$P_1(x)$

1st order approx. at a is $f(a) + (x - a)f'(a)$

$$\lim_{x \rightarrow a} \frac{f(x) - P_1(x)}{x - a} = 0 \quad (\text{The difference should vanish faster than something linear})$$

Check:

$$\lim_{x \rightarrow a} \frac{f(x) - f(a) - (x - a)f'(a)}{x - a} = 0$$

Q: Find second order approximation (polynomial at most quadratic)

Find P_2 s.t.

$$\lim_{x \rightarrow a} \frac{f(x) - P_2(x)}{(x - a)^2} = 0$$

$$\text{Claim: } P_2(x) = f(a) + (x - a)f'(a) + \frac{(x - a)^2}{2} f''(a)$$

Check:

$$\lim_{x \rightarrow a} \frac{f(x) - P_2(x)}{(x-a)^2} = \lim_{x \rightarrow a} \frac{f(x) - f(a) - (x-a)f'(a) - \frac{(x-a)^2}{2} f''(a)}{(x-a)^2}$$

Use L'Hôpital's Rule

Higher dimensional derivatives

$$f: \mathbb{R}^d \rightarrow \mathbb{R}^n$$

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h} \leftarrow \text{Only makes sense if } d=1$$

To solve this, we look at limits along lines

Def

Directional Derivative

$$f: \mathbb{R}^d \rightarrow \mathbb{R}^n$$

$$v \in \mathbb{R}^d, v \neq 0$$

$$D_v f(a) = \left. \frac{d}{dt} f(a+tv) \right|_{t=0}$$
$$= \lim_{t \rightarrow 0} \frac{f(a+tv) - f(a)}{t}$$

Note: $D_v f(a) \in \mathbb{R}^n$

$$D_v f(a) = Df_a v = \frac{\partial}{\partial v} f(a) = \partial_v f(a)$$

Def $i \in \{1, \dots, d\}$ $\partial_{e_i} f(a) =$ i^{th} partial derivative of f at a
 $= \partial_i f(a) = \frac{\partial f}{\partial x_i}(a)$

Example: $f(x) = |x| = \sqrt{\sum_{i=1}^d x_i^2}$

$$\partial_j f = \frac{1}{2\sqrt{\sum_{i=1}^d x_i^2}} \cdot 2x_j = \frac{x_j}{\sqrt{\sum_{i=1}^d x_i^2}} = \frac{x_j}{|x|}$$

"Full Derivative"

$$1D: f \text{ diff at } a \iff \lim_{h \rightarrow 0} \frac{\overbrace{f(a+h) - f(a) - hf'(a)}^{\text{First order approximation}}}{h} = 0$$

What should derivative do in higher dimensions?
Subtract something that vanishes faster than h

In $\mathbb{R}^d$

$f: U \rightarrow \mathbb{R}^n$, U open $U \subseteq \mathbb{R}^d$

Take $a \in U$

We say f is diff at a if

$\exists$ a linear transformation $T: \mathbb{R}^d \rightarrow \mathbb{R}^n$

$$\lim_{h \rightarrow 0} \frac{f(a+h) - f(a) - T(h)}{|h|} = 0$$

Note: Can't possibly have another transformation S
Then $\lim_{h \rightarrow 0} \frac{T(h) - S(h)}{|h|} = 0$ (*)

Def | The derivative of f at a is the unique linear transformation $T: \mathbb{R}^d \rightarrow \mathbb{R}^n$ s.t. (*) holds

Notation: Df_a or $Df(a) = T$

Prp 1 | We say f is diff at $a \iff \exists$ a function e (error) and a L.T. $T: \mathbb{R}^d \rightarrow \mathbb{R}^n$ s.t. $f(a+h) - f(a) - T(h) = e(h)$
and $\lim_{h \rightarrow 0} \frac{e(h)}{|h|} = 0$

Prop 2

If f is diff. at a then $\forall v \in \mathbb{R}^d$, $D_v f(a)$ exists
and $D_v f(a) = (Df_a)(v)$

$$\lim_{t \rightarrow 0} \frac{f(a+tv) - f(a)}{t}$$

L.T. associated with
derivative

Pf

Let T_v be as in Prop 1

Note: $f(a+tv) - f(a)$

$$\begin{aligned} &= f(a+tv) - f(a) - T(tv) + T(tv) \\ &= e(tv) + T(tv) \end{aligned}$$

$$\Rightarrow \lim_{t \rightarrow 0} \frac{f(a+tv) - f(a)}{t} = \lim_{t \rightarrow 0} \left(\frac{e(tv)}{t} + \frac{T(tv)}{t} \right) = 0 + T(v)$$

← Linear

Corollary

f is diff at $a \Rightarrow$

$$Df_a = \begin{pmatrix} | & & | \\ Df_a(e_1) & \dots & Df_a(e_d) \\ | & & | \end{pmatrix} \quad \text{each of these are just the } i\text{th partial}$$

$$= \begin{pmatrix} | & & | \\ \partial_1 f(a) & \dots & \partial_d f(a) \\ | & & | \end{pmatrix}$$

$$= \underbrace{\begin{pmatrix} \partial_1 f(a) & \partial_2 f(a) & \dots & \partial_d f(a) \\ \partial_1 f_2(a) & \vdots & \vdots & \vdots \\ \partial_1 f_n(a) & \partial_2 f_n(a) & \dots & \partial_d f_n(a) \end{pmatrix}}_{\text{Jacobian Matrix}}$$

(First Order Approximation)

Prop All directional derivatives exist $\nRightarrow f$ is diff.

Thm All partials exist AND are cts. at $a \Rightarrow f$ is diff. at a

Recall $f: \mathbb{R}^d \rightarrow \mathbb{R}^n$ is diff. at $a \in \mathbb{R}^d$ if
 $\exists T: \mathbb{R}^d \rightarrow \mathbb{R}^n$ linear and $e: \mathbb{R}^d \rightarrow \mathbb{R}^n$ s.t.
 $f(a+h) = f(a) + Th + e(h)$ and $\lim_{h \rightarrow 0} \frac{e(h)}{|h|} = 0$

Df_a is the unique T from the definition
and
$$Df_a = \begin{pmatrix} \uparrow & & \uparrow \\ \partial_1 f & \cdots & \partial_d f \\ \downarrow & & \downarrow \end{pmatrix}$$
 known as the Jacobian Matrix

Remark f diff. $\Rightarrow$ all partials exist & $Df_a = \text{Jacobian}$
All directional derivatives $\nRightarrow f$ is diff.

Thm All partials exist and cts. at $a \Rightarrow f$ is diff. at a
 $U \subseteq \mathbb{R}^d$ open $a \in U$
 $f: U \rightarrow \mathbb{R}^n$ is cts. and $\partial_i f$ are all cts.
at a

then f is diff. at a

Pf For simplicity let $n=1$ $d=2$

NTS: f is diff. at a

Want: $f(a+h) - f(a) = \text{linear} + \text{smaller than linear}$

$$\begin{aligned} f(a_1+h_1, a_2+h_2) - f(a_1, a_2) &= f(a_1+h_1, a_2+h_2) - f(a_1+h_1, a_2) \\ &\quad + f(a_1+h_1, a_2) - f(a_1, a_2) \end{aligned}$$

By MVT, $\exists \xi_2$ between a_2 and $a_2 + h_2$
 $\exists \xi_1$ between a_1 and $a_1 + h_1$

$$\text{s.t. } f(a+h) - f(a) = h_2 \partial_2 f(a+h_1, \xi_2) + h_1 \partial_1 f(\xi_1, a_2)$$

$$= h_1 \partial_1 f(a) + h_2 \partial_2 f(a) + e(h)$$

$$\text{where } e(h) = h_1 (\partial_1 f(\xi_1, a_2) - \partial_1 f(a)) + h_2 (\partial_2 f(a_1 + h_1, \xi_2) - \partial_2 f(a))$$

Clearly $Th = h_1 \partial_1 f(a) + h_2 \partial_2 f(a)$ is linear in f

$$\text{WTS: } \lim_{h \rightarrow 0} \frac{e(h)}{|h|} = 0$$

$$1. \quad \frac{|h_1 (\partial_1 f(\xi_1, a_2) - \partial_1 f(a))|}{|h|} \leq \underbrace{|\partial_1 f(\xi_1, a_2) - \partial_1 f(a)|}$$

vanishes by continuity of $\partial_1 f$

2. Similarly for the other term

Chain Rule

$$f: \mathbb{R}^n \rightarrow \mathbb{R}^d \quad g: \mathbb{R}^m \rightarrow \mathbb{R}^n$$

Take $a \in \mathbb{R}^m$ with g diff. at a $\leftarrow Dg_a$
 f diff. at $g(a)$ $\leftarrow Df_{g(a)}$

Then $f \circ g$ is diff. at a and

$$D(f \circ g)_a = Df_{g(a)} Dg_a$$

Pf g diff. at $a \Rightarrow \exists e_2$ s.t.

$$g(a+h) = g(a) + Dg_a \cdot h + e_2(h)$$

$$\text{and } \lim_{h \rightarrow 0} \frac{e_2(h)}{|h|} = 0$$

f diff. at $g(a) \Rightarrow \exists e_1$ s.t.

$$f(g(a)+h) = f(g(a)) + Df_{g(a)} \cdot h + e_1(h)$$

$$\text{and } \lim_{h \rightarrow 0} \frac{e_1(h)}{|h|} = 0$$

$$\text{Note: } f(g(a+h)) = f(g(a) + Dg_a \cdot h + e_2(h))$$

$$= f(g(a)) + Df_{g(a)} (Dg_a \cdot h + e_2(h)) + e_1(Dg_a \cdot h + e_2(h))$$

$$= f(g(a)) + Df_{g(a)} \cdot Dg_a \cdot h + e_3(h)$$

$$\text{Where } e_3(h) = Df_{g(a)} \cdot e_2(h) + e_1(Dg_a \cdot h + e_2(h))$$

$$\text{NTS: } \lim_{h \rightarrow 0} \frac{e_3(h)}{|h|} = 0$$

$$\text{Note: } \exists C \in \mathbb{R} \text{ s.t. } \forall v, |Df_{g(a)} v| \leq C|v|$$

$$\forall u, |Dg_a u| \leq C|u|$$

Note: $\frac{|D_{g_2} e_2(h)|}{|h|} \leq \frac{C|e_2(h)|}{|h|} \xrightarrow{h \rightarrow 0} 0$

NTS: $\frac{e_1(D_{g_2} h + e_2(h))}{|h|} \xrightarrow{h \rightarrow 0} 0$

Note: $|D_{g_2} h + e_2(h)| \leq C|h| + \underbrace{|e_2(h)|}_{\leq 1 \cdot |h|}$

By choosing δ small, can ensure
 $0 < |h| < \delta \Rightarrow |e_2(h)| \leq |h|$

So $|D_{g_2} h + e_2(h)| \leq (C+1)|h|$

$\Rightarrow \frac{|e_1(D_{g_2} h + e_2(h))|}{|h|}$ Choose δ so that
 $e_1(v) \leq \epsilon|v|$

$\therefore \frac{e_1(v)}{|v|} \xrightarrow{v \rightarrow 0} 0$

So $\frac{|e_1(v)|}{|h|} \leq \frac{\epsilon|v|}{|h|} \leq (C+1)\epsilon$

Consider the i, j -th coordinate of $D(f \circ g)_a$

$$f: \mathbb{R}^n \rightarrow \mathbb{R}^d$$

$$g: \mathbb{R}^m \rightarrow \mathbb{R}^n$$

$$(D(f \circ g)_a)_{i,j} = \partial_j (f \circ g)_i(a) = \frac{\partial f_i \circ g(a)}{\partial x_j}$$

$$= \sum_{k=1}^n (Df_{g(a)})_{i,k} (Dg(a))_{k,j}$$

$$= \sum_{k=1}^n \frac{\partial f_i(g(a))}{\partial x_k} \cdot \frac{\partial g_k(a)}{\partial x_j} \quad \text{i.e. } \partial_j (f \circ g)_i = \sum_{k=1}^n \underbrace{\partial_k f_i|_g}_{(D_k f)_i \circ g} \cdot \partial_j g_k$$

Say $x_1, \dots, x_m \rightarrow m$ variables

$y_1, \dots, y_n \rightarrow$ some fn of $x_1, \dots, x_m$

$z_1, \dots, z_d \rightarrow$ some fn of $y_1, \dots, y_n$

$$\frac{\partial z_i}{\partial x_j} = \sum_{k=1}^n \frac{\partial z_i}{\partial y_k} \cdot \frac{\partial y_k}{\partial x_j}$$

Higher-Order Derivatives

$$f: \mathbb{R}^d \rightarrow \mathbb{R}$$

Let $k \in \mathbb{N}$ $i_1, i_2, \dots, i_k \in \{1, \dots, d\}$

k -th order partial of f is $\partial_{i_1} \partial_{i_2} \dots \partial_{i_k} f$

Def | $C^k = \{f \mid \text{all partials up to order } k \text{ of } f \text{ exist and are cts.}\}$

Recall | All 1st order partials exist $\nRightarrow f$ diff.

$$f \in C^1 \Rightarrow f \text{ diff.}$$

Q: $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ (all 2nd partials exist)

$$\partial_1 \partial_2 f \stackrel{?}{=} \partial_2 \partial_1 f$$

Note: $\partial_1 f(0) = \lim_{h_1 \rightarrow 0} \frac{f(h_1, 0) - f(0, 0)}{h_1}$

$$\partial_2 \partial_1 f(0) = \lim_{h_2 \rightarrow 0} \frac{\partial_1 f(0, h_2) - \partial_1 f(0, 0)}{h_2}$$

Flipping order might not get the same answer

Thm (Clairaut)

If $f \in C^2$ then $\forall i, j: \partial_i \partial_j f = \partial_j \partial_i f$

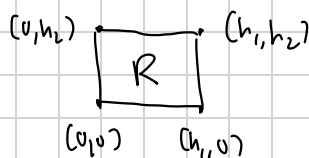
Remark If $\partial_i \partial_j f$ and $\partial_j \partial_i f$ exist in a nbd of a and are cts. at a , then they are equal

Pf Say $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ $a = (0, 0)$

$\partial_1 f, \partial_2 f, \partial_1 \partial_2 f, \partial_2 \partial_1 f$ exist and are cts. at $(0, 0)$

NTS: $\partial_2 \partial_1 f(0) = \partial_1 \partial_2 f(0)$

Let $\Delta(h) = f(0) + f(h_1, h_2) - f(0, h_2) - f(h_1, 0)$



Claim: $\partial_1 \partial_2 f(0) = \lim_{h \rightarrow 0} \frac{\Delta(h)}{h_1 h_2}$

$$\partial_2 \partial_1 f(0) = \lim_{h \rightarrow 0} \frac{\Delta(h)}{h_1 h_2}$$

Note: $\Delta(h) = f(h_1, h_2) - f(h_1, 0) - (f(0, h_2) - f(0, 0))$

Let $g(t) = f(h_1, t) - f(h_1, 0) - f(0, t) + f(0, 0)$

Note: $g(h_2) = \Delta(h)$

$$g(h_2) = g(h_2) - g(0) \stackrel{\text{MVT}}{=} h_2 g'(\xi_2) \text{ for some } \xi_2 \text{ between } 0, h_2$$

$$\text{Note: } g'(\xi_2) = \partial_2 f(h_1, \xi_2) - \partial_2 f(0, \xi_2)$$

$$\stackrel{\text{MVT}}{=} \partial_1 \partial_2 f(\xi_1, \xi_2) \cdot h_1 \text{ for some } \xi_1 \text{ between } 0, h_1$$

Use 1D Calc Theorems when you can

$$\Delta(h) = h_1 h_2 \partial_1 \partial_2 f(\xi_1, \xi_2)$$

$$\text{So } \frac{\Delta(h)}{h_1 h_2} = \underbrace{\partial_1 \partial_2 f(\xi_1, \xi_2)}_{\text{as}}$$

$$\Rightarrow \lim_{h \rightarrow 0} \frac{\Delta(h)}{h_1 h_2} = \partial_1 \partial_2 f(0)$$

$$\text{Similarly } \lim_{h \rightarrow 0} \frac{\Delta h}{h_1 h_2} = \partial_2 \partial_1 f(0) \text{ w/ } f(h_1, h_2) - f(0, h_2) - (f(h_1, 0) - f(0, 0))$$

Thm 1 (Mean Value Theorem)

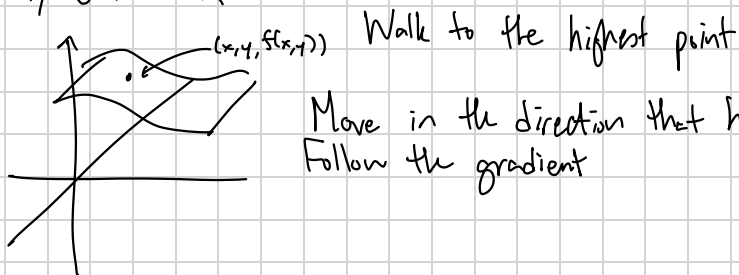
$f: U \rightarrow \mathbb{R}$, $U \subseteq \mathbb{R}^d$ open
 $f \in C^1(U)$, $a, b \in U$ s.t. the line segment between a and b is in U

Then $\exists \xi$ on the line segment s.t.

$$\begin{aligned} f(b) - f(a) &= Df_{\xi}(b-a) \\ &= \nabla f(\xi) \cdot (b-a) \end{aligned} \quad \text{Recall } \nabla f = (Df)^T$$

$$\begin{pmatrix} \partial_1 f \\ \vdots \\ \partial_d f \end{pmatrix}$$

Say $f: \mathbb{R}^2 \rightarrow \mathbb{R}$



What this means is at some point between, when you dot the maximal direction with $(b-a)$ you get $f(b) - f(a)$

$$D_v f(a) = Df_a v = \nabla f(a) \cdot v \leftarrow \text{This maximum will be if } v \text{ is in the direction of the gradient}$$

Pf

$$\begin{aligned} \text{Let } g(t) &= f(a + t(b-a)) & g(1) &= f(b) \\ & & g(0) &= f(a) \\ \text{MVT} \Rightarrow \exists \xi \in (0, 1) \text{ s.t. } & g(1) - g(0) = g'(\xi) \end{aligned}$$

$$\frac{dg}{dt} = \partial_t f(a + t(b-a))$$

$$\begin{aligned}
 \text{Chain Rule } & \sum_{i=1}^d \partial_i f(a + (b-a)) \cdot \frac{d}{dt}(a + (b-a))_i \\
 &= \sum_{i=1}^d \partial_i f(a + (b-a)) (b-a)_i \\
 &= \nabla f(a + (b-a)) \cdot (b-a) \\
 &= \nabla f(a + \xi(b-a)) \cdot (b-a)
 \end{aligned}$$

$$f(b) - f(a) = \nabla f(\xi_2) \cdot (b-a)$$

Taylor's Theorem (2^{nd} order)

$U \subseteq \mathbb{R}^d$ open with $a \in U$, $h \in \mathbb{R}^d$, $f: U \rightarrow \mathbb{R}$, $f \in C^2(U)$

$a, a+h$, & line between the two $\subseteq U$

Then $\exists \xi$ on the line segment joining $a, a+h$ s.t.

$$f(a+h) = f(a) + \nabla f(a) \cdot h + \frac{1}{2} \sum_{i,j=1}^d \partial_i \partial_j f(\xi) h_i h_j$$

Pf

Let $g(t) = f(a+th)$ $g(0) = f(a)$ $g(1) = f(a+h)$

Taylor 1D. $\exists \xi \in (0,1)$ s.t.

$$g(1) = g(0) + g'(0) \cdot (1-0) + \frac{1}{2} g''(\xi) (1-0)^2$$

$$g'(t) = \partial_+ f(a+th) = \sum_i \partial_i f(a+th) h_i$$

$$g''(t) = \partial_+^2 f(a+th) = \sum_i \left(\sum_{j=1}^d \partial_i \partial_j f(a+th) h_j \right) h_i$$

$$f(a+h) = f(a) + \nabla f(a) \cdot h + \frac{1}{2} \sum_{i,j=1}^d \partial_i \partial_j f(a+\xi h) h_i h_j$$

Remark: $f(a+h) = f(a) + \nabla f(a) \cdot h + \frac{1}{2} \sum_{i,j=1}^d \partial_i \partial_j f(a) h_i h_j + R_2(h)$

where $\exists \xi$ on line between $a, a+h$ s.t.

$$R_2(h) = \frac{1}{2} \sum_{i,j=1}^d \partial_i \partial_j (f(\xi) - f(a)) h_i h_j$$

$$\frac{R_2(h)}{|h|^2} \xrightarrow{h \rightarrow 0} 0$$

This is true in $\mathcal{C}^2$

$$\left(\frac{h_i h_j}{|h|^2} \right) \text{ is bounded } \partial_i \partial_j f(\xi) - f(a) \xrightarrow{h \rightarrow 0} 0$$

$$a \rightarrow \xi \rightarrow a$$